

DecodeLabs Industrial Training Kit

Batch 2026

# PROJECT 1

## Data Cleaning & Preparation

*Building the Analysis-Ready Dataset — From Raw Data Audit to Verified Clean Output*

*[ This report was prepared as part of the DecodeLabs Industrial Training Programme, Data Analytics track. It documents the data cleaning and preparation milestone that establishes the verified dataset used in Projects 2–4.]*

### Dataset Analysed

E-Commerce Sales Dataset — 1,200 order records × 14 columns

Order Period: 01 January 2023 — 30 June 2025

Tool: Microsoft Excel

**Submitted to DecodeLabs | July 2026**

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## Executive Summary

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This report documents the complete data cleaning and preparation of a raw e-commerce sales dataset containing 1,200 order records across 14 columns, covering January 2023 to June 2025. The objective was to transform the raw file into a verified, analysis-ready source of truth before any exploratory analysis, SQL work, or dashboard development began.

A full quality audit identified three genuine issues: 309 missing CouponCode values (25.8% of all records), floating-point precision errors affecting every value in the UnitPrice and TotalPrice columns, and inconsistent date display formatting. Missing coupon codes were interpreted as “no coupon used” and imputed with the explicit label NO\_COUPON rather than deleted — preserving all 1,200 records, where listwise deletion would have discarded roughly a quarter of the dataset. Prices were rounded to two decimal places and dates standardised to ISO 8601 (YYYY-MM-DD) format.

Every check and change was recorded in a structured cleaning log (CR001–CR009), with the raw data preserved unchanged in a backup sheet. The cleaned dataset was then re-audited against the project's verification gate — “before you finish, you must prove there are zero duplicate IDs and zero incorrectly formatted dates” — and passed every check with a 0% error rate: 0 duplicate Order IDs, 0 incorrectly formatted dates, 0 blank cells, and 0 calculation mismatches. This verified dataset became the single source of truth for Project 2 (exploratory data analysis), Project 3 (SQL analysis), and Project 4 (Power BI dashboard).

## Section 1: Methodology & Framework

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### 1.1 The Cleaning Workflow

The project follows a four-stage workflow modelled on professional data-engineering practice, in which raw data is never overwritten and every transformation is traceable:

- **Raw Backup:** The original file is copied unchanged into a Raw\_Backup sheet, keeping the process fully reversible.
- **Clean Copy:** All cleaning is carried out on a separate Cleaned\_Dataset sheet, leaving the backup untouched.
- **Documented Log:** Every issue, action, and outcome is recorded in a Log\_Sheet with a unique change ID (CR001–CR009).
- **Verification Gate:** The cleaned data is re-audited against a fixed set of pass/fail checks; the project is complete only when every check returns zero errors.

This mirrors the Gatekeeper principle: no analysis is performed on data that has not passed verification. A dataset that fails any check is returned to the cleaning stage, not partially accepted.

### 1.2 Data Source & Scope

Source: Dataset\_for\_Data\_Analytics.xlsx, provided by DecodeLabs as part of the Data Analytics track.

Scope: 1,200 e-commerce order records × 14 columns — OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, and TotalPrice — spanning 1,189 unique customers, 7 product categories, 5 payment methods, 5 order statuses, and 5 referral sources across January 2023 to June 2025.

### 1.3 Key Techniques Applied

- **Missing-value audit:** column-by-column null count across all 14 fields.
- **Duplicate detection:** Remove Duplicates across all columns, plus a COUNTIF audit on the OrderID unique identifier.
- **Numeric precision correction:** ROUND to 2 decimal places with a consistent 0.00 custom number format.
- **Date standardisation:** type confirmation (true Excel dates, not text) and ISO 8601 (YYYY-MM-DD) display format.
- **Cross-field validation:** recomputing Quantity × UnitPrice and reconciling against TotalPrice to a 0.01 tolerance.
- **Category & text hygiene audits:** distinct-value inventories on categorical columns and whitespace/case checks on text fields.

## Section 2: Step 1 — Data Quality Audit

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Before any changes were made, the raw dataset was audited column by column for the five failure modes the project brief targets: missing or null values, duplicate records, incorrect formats, calculation errors, and category inconsistencies. Auditing first — rather than fixing issues as they are noticed — ensures the full scope of the problem is known before any cell is edited, and gives the cleaning log a complete baseline.

The audit surfaced three genuine issues: 309 missing `CouponCode` values (25.8% of records — the only column with missing data; all other 13 columns are fully populated), floating-point precision artefacts in both price columns (e.g. 533.8099999999999 stored in place of 533.81), and inconsistent date display formatting. The remaining checks — duplicates, cross-field integrity, category consistency, and text hygiene — returned zero findings, and served as verification that no hidden problems existed in the raw file.

## Section 3: Step 2 — Missing Value Treatment (`CouponCode`)

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The 309 blank `CouponCode` cells represent 25.8% of the dataset — far too large a share to delete. Listwise deletion would have discarded roughly one in four orders and biased every downstream analysis toward coupon-using customers. Instead, the blanks were interpreted through business logic: an empty coupon field on an otherwise complete order record means no coupon was used at checkout, not that the data is unknown.

Each blank was therefore filled with the explicit label `NO_COUPON`, converting an ambiguous gap into an analysable category and preserving all 1,200 records.

This decision paid off directly in later projects: coupon vs non-coupon comparisons became a standard cut in the Project 2 EDA and a dedicated visual in the Project 4 Power BI dashboard — neither of which would have been possible had the blank rows been deleted or left as nulls.

## Section 4: Step 3 — Numeric Precision Correction

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Every value in the UnitPrice and TotalPrice columns (1,200 cells each) carried floating-point representation artefacts — values such as 533.8099999999999 and 147.41999999999999 stored in place of their true 2-decimal currency amounts. Left uncorrected, these artefacts cause three practical problems: misleading display values, string-comparison failures when data is exported to other tools, and micro-discrepancies that accumulate in aggregations.

Both columns were corrected in two layers: the stored values were rounded to exactly 2 decimal places using ROUND, and a consistent 0.00 custom number format was applied so every price displays at uniform currency precision. Correcting the stored value — not just the display format — matters because the dataset was destined for import into SQL Server (Project 3), where display formatting does not travel with the data.

## Section 5: Step 4 — Date Standardisation

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The Date column was audited in two stages. First, every value was confirmed to be a true Excel date serial rather than text masquerading as a date — the failure mode that silently breaks sorting, filtering, and time-series grouping. Second, the ISO 8601 display format (YYYY-MM-DD) was applied across all 1,200 values.

ISO 8601 was chosen deliberately: it is unambiguous across regional conventions (avoiding the UK/US day-month reversal problem), sorts correctly as text, and is the native date format of SQL Server and Power BI — the two tools this dataset feeds in Projects 3 and 4. The audit confirmed 0 invalid dates and a 0% error rate on date formats, directly satisfying the second half of the project's verification requirement.

## Section 6: Step 5 — Duplicate & Integrity Verification

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Four further audits confirmed the structural integrity of the dataset:

- **Full-row duplicates:** Remove Duplicates across all 14 columns found 0 fully duplicated records.
- **Duplicate identifiers:** a COUNTIF audit on OrderID confirmed all 1,200 identifiers are unique — 0 duplicate IDs, satisfying the first half of the verification requirement.
- **Cross-field integrity:** Quantity × UnitPrice was recomputed for every record and reconciled against the stored TotalPrice to a 0.01 tolerance — 0 mismatches found.
- **Category & text hygiene:** distinct-value inventories confirmed exactly 7 products, 5 payment methods, 5 order statuses, and 5 referral sources, with no spelling or case variants; whitespace and case checks on all text columns found no issues.

## Section 7: Cleaning Log

Every check and change was recorded in the Log\_Sheet with a unique change ID, the records affected, the action taken, and the outcome. Items marked Resolved involved a change to the data; items marked Verified were audits that confirmed no change was needed.

| ID    | Issue Type              | Column(s)                           | Records | Action Taken                                                                                         | Impact / Result                   | Status   |
|-------|-------------------------|-------------------------------------|---------|------------------------------------------------------------------------------------------------------|-----------------------------------|----------|
| CR001 | Missing values          | CouponCode                          | 309     | Blanks interpreted as “no coupon used” and filled with the label NO_COUPON instead of deleting rows. | All 1,200 records preserved       | Resolved |
| CR002 | Duplicate full rows     | All columns                         | 0       | Checked with Remove Duplicates across all 14 columns.                                                | 0 duplicates found; none removed  | Verified |
| CR003 | Duplicate identifiers   | OrderID                             | 0       | COUNTIF audit on OrderID.                                                                            | 0 duplicate IDs — 0% error rate   | Verified |
| CR004 | Incorrect number format | UnitPrice                           | 1,200   | Rounded to 2 dp (floating-point artefacts, e.g. 533.8099999999999) and applied 0.00 format.          | All prices at 2-decimal precision | Resolved |
| CR005 | Incorrect number format | TotalPrice                          | 1,200   | Rounded to 2 dp and applied 0.00 format.                                                             | All totals at 2-decimal precision | Resolved |
| CR006 | Date standardisation    | Date                                | 1,200   | Confirmed all values are true dates; applied ISO 8601 format (YYYY-MM-DD).                           | 0 invalid dates                   | Verified |
| CR007 | Cross-field integrity   | TotalPrice vs Qty × UnitPrice       | 0       | Recomputed Quantity × UnitPrice vs TotalPrice (tolerance 0.01).                                      | 0 mismatches; totals reconciled   | Verified |
| CR008 | Category consistency    | Product / Payment / Status / Source | 0       | Distinct-value audit: 7 products, 5 payment methods, 5 statuses, 5 sources.                          | No spelling or case variants      | Verified |
| CR009 | Text hygiene            | All text columns                    | 0       | Checked leading/trailing whitespace and case inconsistencies.                                        | None found                        | Verified |

## Section 8: Verification Gate Results

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The project brief sets an explicit completion standard: “Before you finish, you must prove there are zero duplicate IDs and zero incorrectly formatted dates.” The cleaned dataset was re-audited against this gate, extended with two further checks covering completeness and calculation integrity. All four checks passed with a 0% error rate:

| Verification Check                                  | Result | Status |
|-----------------------------------------------------|--------|--------|
| Duplicate Order IDs                                 | 0      | PASS   |
| Incorrectly formatted dates                         | 0      | PASS   |
| Blank / missing cells                               | 0      | PASS   |
| TotalPrice calculation mismatches (Qty × UnitPrice) | 0      | PASS   |

The dataset therefore passes the Project 1 verification gate in full and qualifies as the source of truth for all subsequent projects in the track.

## Section 9: Conclusion

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Project 1 delivered a verified, analysis-ready dataset with zero data loss: all 1,200 records were retained, with the single missing-data problem solved by business-logic imputation rather than deletion. The raw file remains preserved in its backup sheet, every transformation is documented against a change ID, and the final output passed all verification checks with a 0% error rate.

The discipline established here — audit first, never overwrite raw data, log every change, and gate completion on verification — carried directly into the rest of the track: the Project 2 EDA figures, the Project 3 SQL cross-validation checks, and the Project 4 Power BI measures all reconcile back to this cleaned dataset. Data integrity, not visualisation, is the foundation the remaining projects are built on.

## Summary Results Table

The table below consolidates every key metric from this project into a single reference.

| Metric                                   | Result                                             |
|------------------------------------------|----------------------------------------------------|
| Records (raw → cleaned)                  | 1,200 → 1,200 (0 deleted)                          |
| Columns                                  | 14                                                 |
| Order period                             | 01 Jan 2023 – 30 Jun 2025                          |
| Unique customers                         | 1,189                                              |
| Issues identified                        | 3 (missing values, numeric precision, date format) |
| Missing CouponCode values treated        | 309 (25.8%) → imputed as NO_COUPON                 |
| Price values corrected to 2 dp           | 2,400 (UnitPrice + TotalPrice)                     |
| Dates standardised to ISO 8601           | 1,200                                              |
| Duplicate full rows / duplicate OrderIDs | 0 / 0                                              |
| TotalPrice calculation mismatches        | 0                                                  |
| Cleaning log entries                     | 9 (CR001–CR009: 3 Resolved, 6 Verified)            |
| Verification gate result                 | PASS — 0% error rate on all checks                 |
| Tool                                     | Microsoft Excel                                    |

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